

period of the function $f(x) = ax \bmod N$, where $N=15$ and 'a' is a randomly chosen integer coprime to N .

Design approach: The algorithm consists of quantum modular exponentiation, quantum Fourier transform, and classical post-processing. In practice, quantum circuits for Shor's algorithm look like what's shown in Figure 1.

```
# Skeleton of Shor's Algorithm in Qiskit
from qiskit.algorithms import Shor
result = Shor().factor(N=15)
print("Factors of 15:", result.factors)
```

While this example is simplified, actual implementations require intricate circuit design and error correction, especially for larger numbers.

Grover's algorithm: Speeding up search problems

Grover's algorithm, devised by Lov Grover in 1996, addresses the challenge of searching unsorted databases. Classically, searching N items requires $O(N)$ time, but Grover's algorithm achieves this in $O(\sqrt{N})$ time, providing a quadratic speedup. The algorithm leverages quantum amplitude amplification, repeatedly applying an oracle and a diffusion operator to increase the probability of

finding the desired item. Figure 2 gives a reference architecture of Grover's algorithm to visually understand the design approach.

Example: Database search: Suppose we need to locate a specific entry in a database of 100,000 items. Classically, this would require, on average, 50,000 checks. Grover's algorithm can accomplish this with only about 317 queries.

Here's the code snippet:

```
# Grover's Algorithm in Qiskit
from qiskit import QuantumCircuit
from qiskit.circuit.library import GroverOperator
oracle = QuantumCircuit(2)
oracle.cz(0, 1) # Example oracle
grover_op = GroverOperator(oracle)
```

Grover's algorithm is versatile, applicable to various search and optimisation problems, and forms the basis for quantum-enhanced combinatorial optimisation.

Recent breakthroughs: New algorithms on the horizon

The quantum computing landscape is rapidly evolving, with new algorithms emerging to tackle optimisation, simulation, and machine learning problems. Hybrid quantum-classical

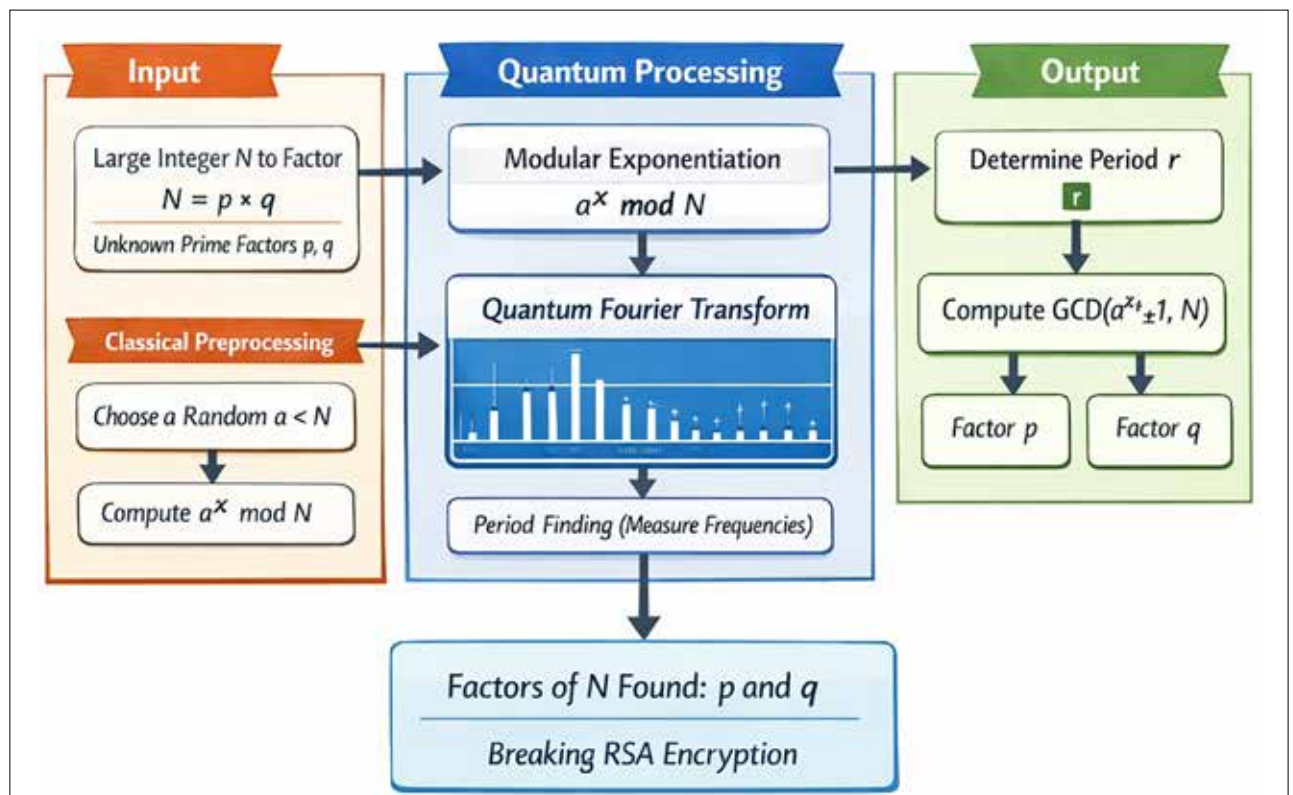


Figure 1: Shor's quantum algorithm for integer factorisation